DO NOW — RECALL

Lesson 119 · Spiral Development and Method Selection

Name: _____ Class: _____ Date: _____

1. What is one distinguishing XP practice?

2. An OR gate has inputs 0 and 1. What is the output?

3. What is the effect of enqueue(X), enqueue(Y), dequeue() on an initially empty FIFO queue?

4. What is the time complexity of sequential search in the worst case?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

For example TDD, pair programming, CI or frequent refactoring.

2. Model answer

1.

3. Model answer

X is removed; Y remains at the front.

4. Model answer

O(n).

By the end, you can…

• Set objectives, options and constraints for a spiral iteration.

• Evaluate and reduce technical or organisational uncertainties before commitment.

• Develop/verify increment, consult stakeholders, plan next risk-driven cycle.

Vocabulary — know these terms

spiral model — Risk-driven iterative development lifecycle.

risk analysis — Estimating likelihood and impact of potential problems.

prototype — Early model built to explore or test an idea.

iteration — Repeated cycle of development and evaluation.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Work through the method

Use the middle stage of the visual to follow the process and state each intermediate result.

Shared worked answer

RAD/agile could produce accessible prototypes quickly with visitor feedback, but test accessibility systematically. Spiral may be excessive for a small low-risk app; compare project scale and uncertainty.

IMPROVE: Explain one correction to your own reasoning.

NEW LEARNING

Read, annotate and explain

Plan the cycle

Set objectives, options and constraints for a spiral iteration.

Analyse risks

Evaluate and reduce technical or organisational uncertainties before commitment.

Prototype and review

Develop/verify increment, consult stakeholders, plan next risk-driven cycle.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

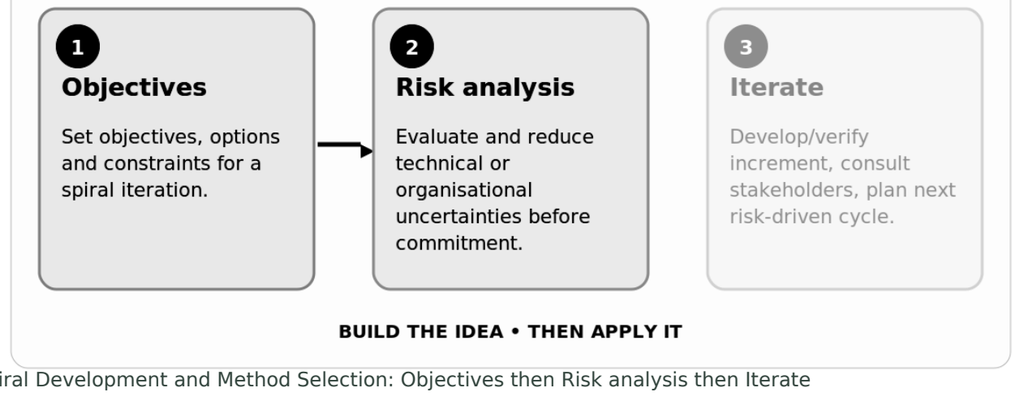
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Spiral Development and Method Selection: Objectives then Risk analysis then Iterate

New Learning



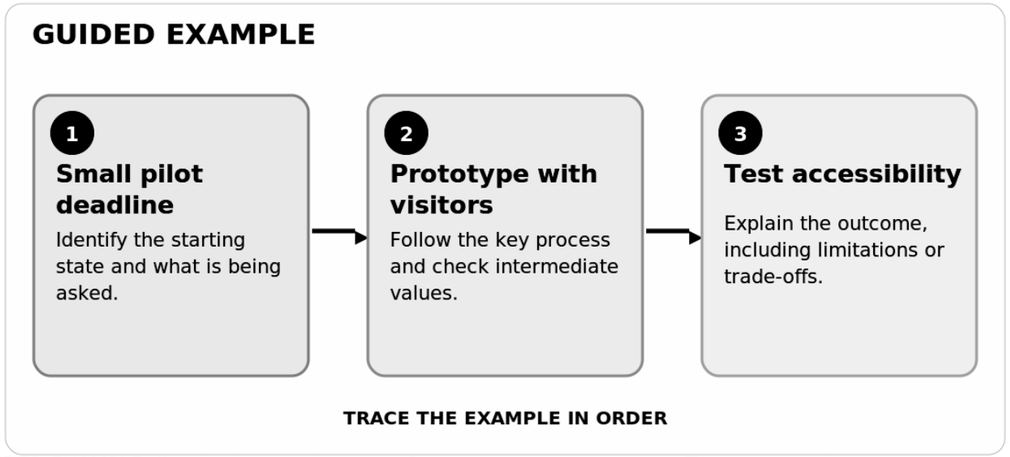
Spiral Development and Method Selection: Objectives then Risk analysis then Iterate

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A museum app needs a basic pilot within weeks but must support disabled visitors. Select a method.

Identify the inputs, relevant facts and exact output or judgement required.



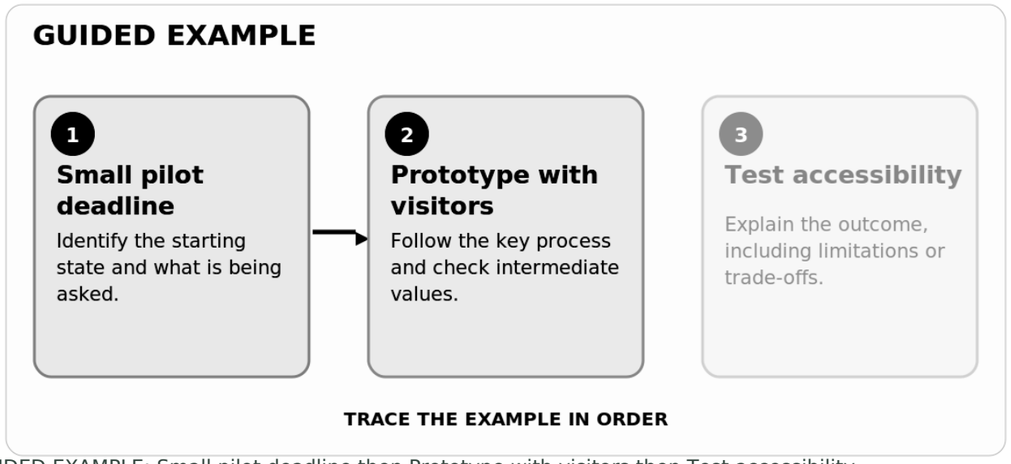
GUIDED EXAMPLE: Small pilot deadline then Prototype with visitors then Test accessibility

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A museum app needs a basic pilot within weeks but must support disabled visitors. Select a method.

Identify the inputs, relevant facts and exact output or judgement required.

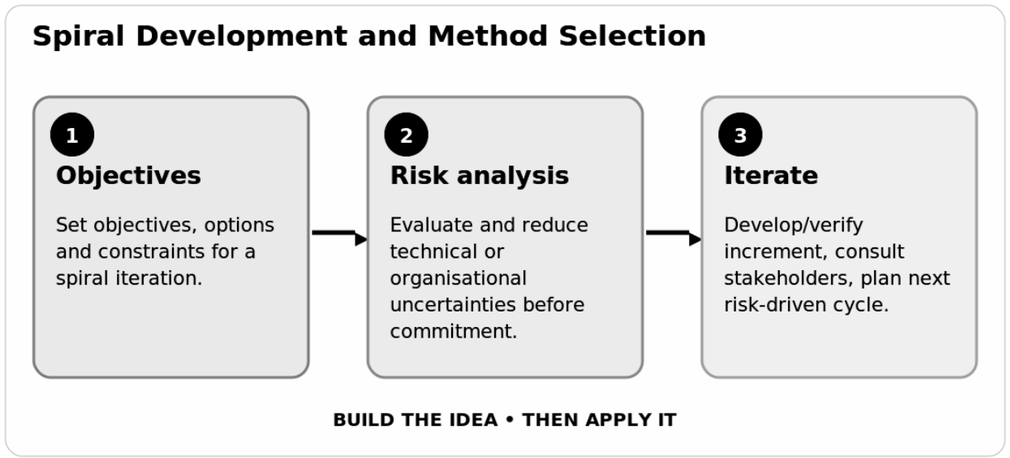


GUIDED EXAMPLE: Small pilot deadline then Prototype with visitors then Test accessibility

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Spiral Development and Method Selection: Objectives then Risk analysis then Iterate

I DO — WORKED EXAMPLE

Follow the worked method

A bank is building a novel payment verification system with security uncertainty. Justify the spiral model.

Model answer

Each cycle sets goals, identifies risks such as fraud or performance, prototypes/assesses the riskiest elements, evaluates with stakeholders then plans the next cycle. Risk reduction drives the project.

Check: link each claim to the scenario and use precise OCR terminology.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Spiral development is simply waterfall repeated without evaluating project risks.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

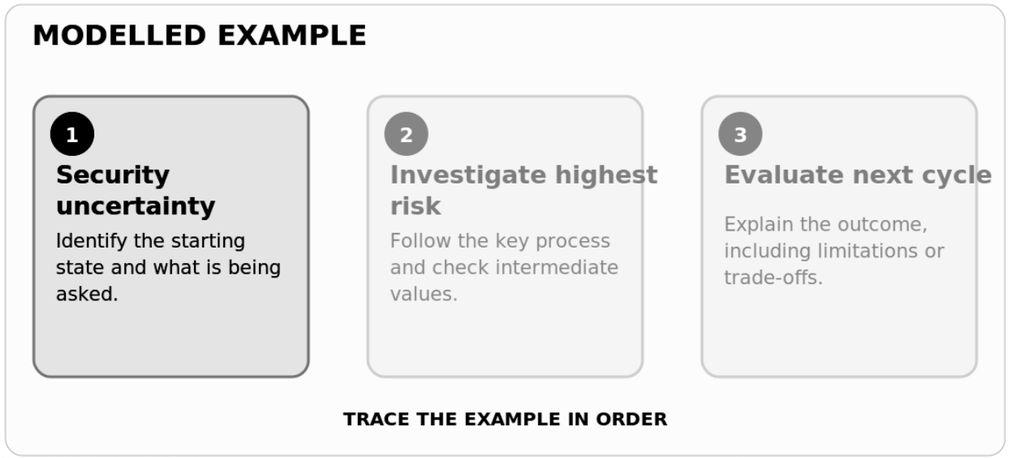
Compare waterfall, agile/RAD, XP and spiral for a high-risk changing specification. [8]

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A bank is building a novel payment verification system with security uncertainty. Justify the spiral model.

Each cycle sets goals, identifies risks such as fraud or performance, prototypes/assesses the riskiest elements, evaluates with stakeholders then plans the next cycle. Risk reduction drives the projec



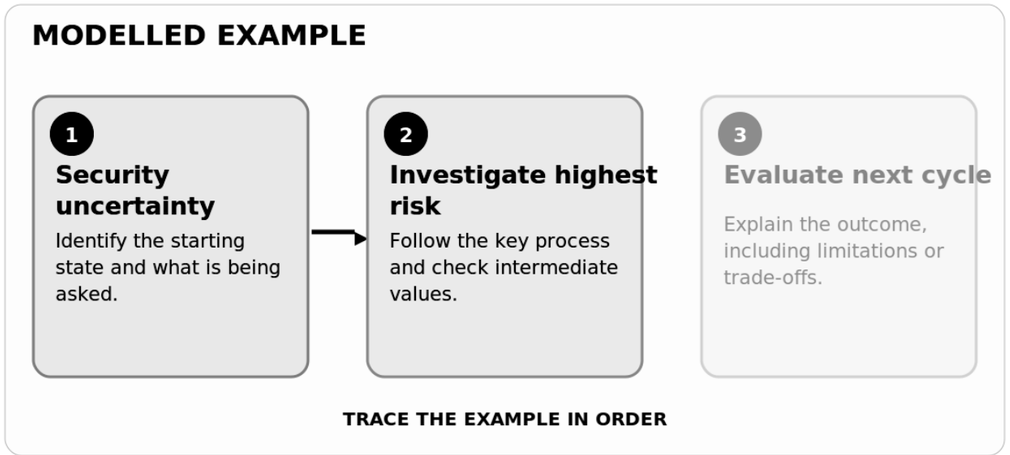
MODELLED EXAMPLE: Security uncertainty then Investigate highest risk then Evaluate next cycle

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A bank is building a novel payment verification system with security uncertainty. Justify the spiral model.

Each cycle sets goals, identifies risks such as fraud or performance, prototypes/assesses the riskiest elements, evaluates with stakeholders then plans the next cycle. Risk reduction drives the projec



MODELLED EXAMPLE: Security uncertainty then Investigate highest risk then Evaluate next cycle

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Describe three central activities within one spiral development cycle. [6]

Question 2

Give two reasons a fixed-requirement small project might not benefit from the spiral model. [6]

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: What must we decide? Identify the inputs, relevant facts and exact output or judgement required.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

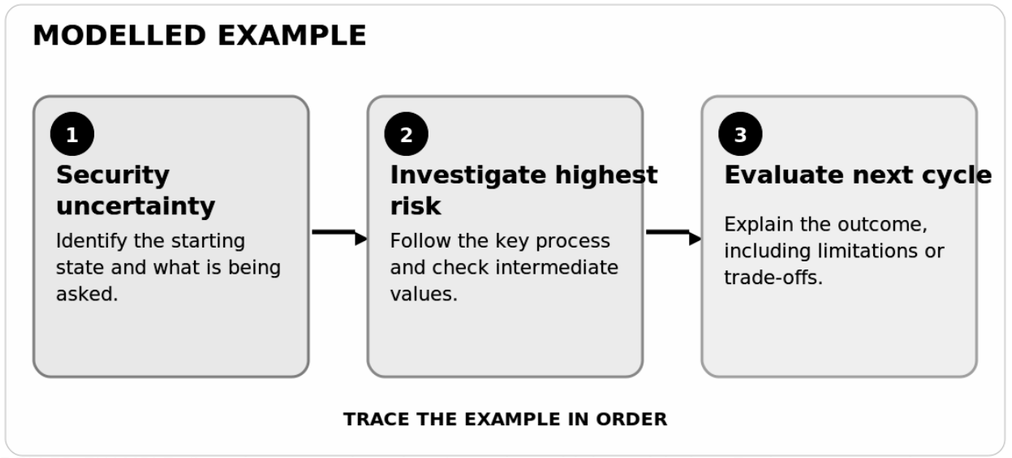
IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A bank is building a novel payment verification system with security uncertainty. Justify the spiral model.

Each cycle sets goals, identifies risks such as fraud or performance, prototypes/assesses the riskiest elements, evaluates with stakeholders then plans the next cycle. Risk reduction drives the projec



MODELLED EXAMPLE: Security uncertainty then Investigate highest risk then Evaluate next cycle

WE DO — GUIDED PRACTICE

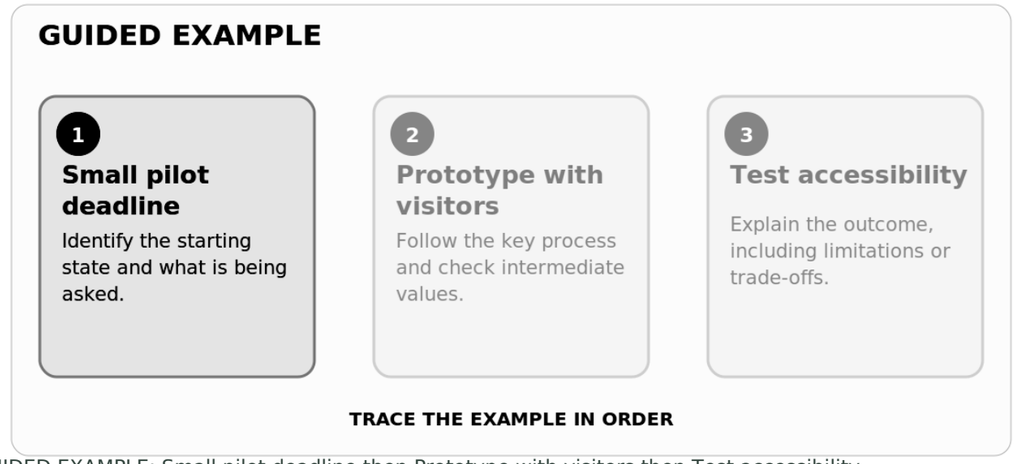
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A museum app needs a basic pilot within weeks but must support disabled visitors. Select a method.

Identify the inputs, relevant facts and exact output or judgement required.



GUIDED EXAMPLE: Small pilot deadline then Prototype with visitors then Test accessibility